KONE ELEVATE — RESEARCH DOCUMENTATION

KONE Elevate — Complete Five-Subsystem Fault Dictionary

Drive / Motor · Door System · Brake / Traction
Controller / Safety-Circuit · Sensors / Instrumentation

Reference taxonomy for the Autonomous Fault Isolation and Root Cause Analysis knowledge base

Team RiseForge · KONE Elevate Hackathon

Document Overview

This dictionary organizes elevator diagnostic knowledge into **five subsystems**. Each subsystem lists components, sensors/observable data, failure modes, and signal-to-root-cause mappings. These are **candidate hypotheses** for the RCA engine — not confirmed diagnoses. Live reasoning must evaluate correlated evidence before promoting any candidate to a diagnosis.

Subsystem	Scope
1. Drive / Motor	Power conversion, drive, motor, motion generation
2. Door System	Door operator, sensors, lock/interlock, cycle mechanics
3. Brake / Traction	Brake, sheave, ropes/belts, governor, safety gear
4. Controller / Safety-Circuit	PLC, I/O, safety chain, relays, event logging
5. Sensors / Instrumentation	Measurement layer providing RCA evidence

RCA Principle

Overcurrent is a **symptom**, not a root cause. The same fault code can have multiple upstream causes. The RCA engine needs **correlated evidence across subsystems**, not a simple lookup.

1. DRIVE / MOTOR

This subsystem covers the electrical power conversion, drive, motor, and motion-generation chain. Relevant components include the VFD/VVVF, DC bus, inverter, PWM/IGBT stage, gate drivers, current/voltage sensing, regenerative braking, PMSM, encoder feedback, motor current and motor temperature.

Fundamental motion chain: ***motor*** → ***drive*** → ***traction sheave*** → ***ropes/belts***

Components / Devices / Equipment

- Traction motor
- PMSM — Permanent Magnet Synchronous Motor
- Motor stator
- Motor rotor
- VFD / VVVF drive
- AC input stage
- Rectifier
- DC bus
- DC-link capacitors
- Inverter stage
- IGBT power modules
- IGBT gate drivers
- PWM switching circuitry
- Braking chopper
- Braking resistor
- Motor current sensors
- Drive voltage sensors
- DC-bus voltage sensing
- Drive temperature sensing
- Motor temperature sensing
- Motor encoder / feedback interface
- Motor-control electronics
- Torque-control circuitry
- Speed-control circuitry
- Regenerative-braking circuitry
- Motor cables
- Power cables
- Electrical terminals/connections

Sensors / Observable Data

- Phase current
- Motor current
- Motor voltage
- DC-bus voltage
- Drive temperature
- Motor temperature
- Motor speed
- Encoder feedback
- Torque
- Power-quality measurements
- Vibration
- Acceleration
- Overcurrent alarm
- Overvoltage alarm
- Undervoltage alarm
- Overload alarm
- Drive fault code
- Motor fault indication

Possible Failure Modes

1. IGBT short circuit
2. IGBT thermal degradation
3. DC-bus overvoltage
4. DC-bus undervoltage
5. DC-link capacitor degradation
6. Motor overcurrent
7. Motor overload
8. Motor winding phase-to-phase fault
9. Motor winding ground fault
10. Motor insulation degradation
11. Mechanical jam
12. Seized bearing
13. Excessive mechanical load
14. Incorrect motor parameters
15. Drive/inverter failure
16. Gate-driver fault

17. Current-sensor fault
18. Voltage-sensor fault
19. Regenerative-braking abnormality
20. Motor overheating
21. Encoder feedback problem
22. Motor cable fault
23. Power-supply disturbance

> **Key RCA Insight**
> Overcurrent is a **symptom rather than a root cause**. It can originate from mechanical binding, motor/cable electrical faults, an internal IGBT failure, or incorrect VFD parameters.

Observable Signals → Plausible Root Causes

Observable Signal	Possible Root Cause
Sudden very high motor current	Mechanical jam / seized bearing
Sudden overcurrent + drive fault	IGBT failure, winding fault, mechanical jam
Repeated IGBT temperature spikes	Thermal stress / cooling problem
High DC-bus voltage during deceleration	Regenerative-energy problem
DC-bus voltage instability	DC-link capacitor degradation / power issue
Phase-current imbalance	Motor winding fault / cable fault / drive fault
High current with mechanical binding	Seized bearing / mechanical obstruction
High current without corresponding mechanical load	Electrical motor fault / IGBT fault
Motor temperature gradually increasing	Overload / cooling degradation / bearing friction
Drive faults even when motor-side is isolated	Internal drive / IGBT fault
Abnormal current waveform	Motor winding / rotor / bearing / drive problem
Motor speed deviation	Encoder / drive / motor-control problem
Overcurrent + brake timing violation	Brake/mechanical-load problem (not necessarily electrical)

The last example is critical: the same fault code can have multiple upstream causes — which is why the RCA engine needs correlated evidence rather than a simple lookup.

2. DOOR SYSTEM

The door system is a major diagnostic subsystem. Relevant components include the door operator, door motor, belt, rollers, tracks, lock/interlock, photo-eye, light curtain, safety edge, encoder, and door-position/limit sensing.

Components / Devices / Equipment

- Door operator
- Door motor
- Door drive/controller
- Door belt
- Door chain (where applicable)
- Door rollers
- Door bearings
- Door tracks
- Door panels
- Door lock
- Door-lock contacts
- Door interlock
- Photo-eye
- Light curtain
- Safety edge
- Door encoder
- Open-position limit sensor
- Close-position limit sensor
- Door-zone sensing
- Door-position sensing
- Door motor/current sensing
- Door-control electronics
- Door wiring/connectors

Sensors / Observable Data

- Door position
- Door opening time
- Door closing time
- Door-cycle duration
- Door motor current
- Door motor load
- Door encoder position
- Door encoder speed
- Door-lock state
- Interlock state
- Photo-eye state
- Light-curtain state
- Safety-edge state
- Open-limit state
- Close-limit state
- Number of reopening events
- Door timeout
- Door fault alarm

Possible Failure Modes

1. Roller wear
2. Roller bearing wear
3. Belt wear/stretch
4. Track misalignment
5. Door mechanical obstruction
6. Door motor degradation
7. Door encoder drift
8. Door encoder failure
9. Photo-eye contamination
10. Photo-eye misalignment
11. Light-curtain degradation
12. Safety-edge fault
13. Door-lock contact contamination
14. Door-lock contact pitting/wear
15. Door interlock fault
16. Open-limit sensor fault
17. Close-limit sensor fault
18. Door wiring/connector fault
19. Door-control electronics fault
20. Excessive door closing force

21. Door-cycle-time degradation

Observable Signals → Plausible Root Causes

Observable Signal	Plausible Root Cause
Increasing door-cycle time	Roller/bearing wear
Increasing door motor current	Roller friction / belt wear / mechanical obstruction
Slow or noisy door movement	Roller/track/belt degradation
Repeated door reopening	Photo-eye / light-curtain / obstruction
Reopening despite no physical obstruction	Sensor drift / contamination
Unstable "door closed" signal	Door-lock contact contamination/pitting
Door-lock failure alarm	Lock/interlock problem
Position deviation	Door encoder drift
Door stops before fully opening	Open-limit / encoder / operator problem
Door stops before fully closing	Close-limit / lock / sensor problem
Door timeout	Mechanical wear / motor degradation / obstruction
Door current increasing over many cycles	Progressive mechanical wear
Intermittent door fault	Wiring / contact / encoder degradation

Documented Causal Chains

- **Roller wear** → slow/noisy door → increased current → door timeout
- **Photo-eye drift** → reopening → abnormal sensor state → door fault
- **Encoder drift** → incorrect position → position deviation → encoder fault
- **Lock-contact problem** → repeated reopening → unstable lock state → door-lock fault

3. BRAKE / TRACTION

This combined subsystem covers the braking mechanism, traction sheave, ropes/belts, rope tension, and traction-related safety mechanisms: electromechanical brake, brake coil, brake wear, torque, timing, monitoring/self-test, steel ropes/coated belts, traction sheave, rope tension/wear, sheave wear, rope slip, counterweight imbalance, overspeed governor, safety gear, UCMP/UCM, ACOP, and buffers.

Components — Brake

- Electromechanical brake
- Brake coil
- Brake armature/linkage
- Brake friction surfaces/lining
- Brake monitoring mechanism
- Brake self-test mechanism
- Brake torque mechanism

Components — Traction

- Traction sheave
- Steel suspension ropes
- Coated belts (where applicable)
- Rope termination
- Rope tensioning system
- Counterweight
- Counterweight-related mechanical load
- Machine/traction assembly

Components — Safety-Related

- Overspeed governor
- Governor rope
- Safety gear
- UCMP/UCM protection
- ACOP-related protection
- Buffers

Sensors / Observable Data

- Brake timing
- Brake release timing
- Brake application timing
- Brake torque
- Brake self-test result
- Rope tension
- Rope condition
- Wire-break indicators/count
- Traction slip
- Motor speed
- Car speed
- Position
- Vibration
- Acceleration
- Unexpected movement
- Overspeed condition
- UCM/UCMP event
- Brake fault alarm
- Rope-condition alarm

Possible Failure Modes

1. Brake torque below specification
2. Brake-torque miscalibration
3. Brake wear
4. Brake friction degradation
5. Brake coil problem

6. Brake-release timing problem
7. Brake-application timing problem
8. Brake monitoring fault
9. Brake self-test failure
10. Rope wear
11. Broken wire strands
12. Rope corrosion
13. Rope diameter degradation
14. Unequal rope tension
15. Rope slip
16. Traction loss
17. Traction-sheave wear
18. Sheave contamination
19. Counterweight imbalance
20. Overspeed condition
21. UCM/UCMP event
22. Governor-related fault
23. Safety-gear activation
24. Buffer-related abnormality

Observable Signals → Plausible Root Causes

Observable Signal	Plausible Root Cause
Brake timing violation	Brake mechanism / timing problem
Brake slip	Insufficient brake torque
Unexpected car movement	Brake / traction problem
Holding fault	Insufficient brake torque / brake wear
Brake torque below expected value	Brake wear / adjustment / calibration
Increasing rope vibration	Rope degradation / traction problem
Rope-condition alarm	Rope wear / corrosion / broken wires
Increasing wire-break count	Rope degradation
Slip under load	Sheave wear / insufficient traction / rope tension issue
Speed deviation under load	Traction slip
Unequal rope tension	Rope / tensioning problem
Excessive vibration	Sheave / rope / traction degradation
Overspeed event	Overspeed protection / drive / traction abnormality
UCM/UCMP event	Unexpected car movement requiring safety response
Motor current ↑ while brake not fully released	Brake drag / mechanical resistance

Cross-Subsystem RCA Distinction

A motor can appear abnormal because the **brake/traction subsystem is imposing excessive mechanical load**. That is exactly the kind of cross-subsystem reasoning this project is designed to perform.

4. CONTROLLER / SAFETY-CIRCUIT

This subsystem covers the decision/control electronics and the independent safety chain: PLC/controller, main control board, I/O modules, safety controller, drive interface, communication buses, event logging, fault-code generation, and state-machine logic.

Components — Controller

- Elevator PLC/controller
- Main control board
- CPU/control processor
- I/O modules
- Digital inputs
- Digital outputs
- Analog inputs
- Analog outputs (where applicable)
- Safety controller
- Drive interface
- Communication interfaces/buses
- Controller power supply
- Control wiring
- Relays
- Contactors
- Relay coils
- Terminal blocks/connectors

Components — Safety Circuit

- Safety chain
- Door interlock contacts
- Governor safety contacts
- Emergency-stop contacts
- Final-limit/safety contacts (where applicable)
- Safety relays
- Safety contactors
- Safety controller/interface
- Hardwired safety path

Sensors / Observable Data

- Safety-chain state
- Door-interlock state
- Emergency-stop state
- Governor state
- Relay state
- Contactor state
- Controller state
- I/O state
- Communication status
- Controller event logs
- Fault logs
- Fault codes
- Warning codes
- Hard-fault status
- Soft-fault status
- Reset events
- State-machine transitions
- Drive communication status

Possible Failure Modes

1. Safety-chain open
2. Door-interlock contact open
3. Governor contact open
4. Emergency-stop activated
5. Final-limit contact open
6. Relay-coil degradation
7. Contact contamination
8. Contact pitting/wear
9. Loose terminal/wiring
10. Controller I/O failure
11. Controller main-board fault

12. PLC logic fault
13. Communication fault
14. Drive-controller communication failure
15. Controller power-supply problem
16. Firmware/software issue
17. I/O module failure
18. Intermittent controller fault
19. State-machine/logic abnormality
20. False/intermittent safety-chain indication

Observable Signals → Plausible Root Causes

Observable Signal	Plausible Root Cause
Safety circuit open	Open safety contact
Elevator immediately held out of service	Safety-chain interruption
Door interlock open	Door-lock / interlock problem
Governor contact open	Governor / safety event
Emergency-stop active	Emergency-stop activation
Intermittent safety fault	Relay / contact degradation
Repeated reset required	Intermittent contact / relay problem
Multiple unexplained alarms	Controller / I/O / communication problem
Erratic I/O behavior	I/O module or wiring fault
Communication loss with drive	Communication / interface fault
Controller reboot	Power-supply / control-board problem
Fault appears/disappears intermittently	Relay / contact / wiring degradation
Controller reports several dependent alarms	Primary controller/safety event causing downstream alarms
Hard fault remains after reset	Persistent safety-chain / component failure

Safety Boundary (Non-Negotiable)

The AI is **not part of the physical safety-control loop**. The elevator's hardwired safety chain remains responsible for safety-critical stopping behavior; the AI diagnoses and explains what happened. This distinction must remain explicit in the architecture.

5. SENSORS / INSTRUMENTATION

This is the broad measurement and telemetry layer that provides evidence to the RCA engine. It covers electrical, mechanical, positional, door, environmental, and operational measurements.

Electrical

- Motor-current sensor
- Phase-current sensors
- Motor-voltage sensing
- DC-bus-voltage sensor
- Drive-voltage sensing
- Power-quality measurement
- Drive-temperature sensor
- Motor-temperature sensor

Mechanical

- Vibration sensor
- Accelerometer
- Brake-torque measurement
- Rope-tension measurement
- Motor-speed measurement

Position / Motion

- Rotary encoder
- Absolute encoder
- Position sensor
- Car-position sensing
- Floor-leveling sensor
- Door encoder
- Door-position sensor

Door

- Door-position sensor
- Door-cycle timer/data
- Door-motor-current sensing
- Door-lock-state sensing
- Photo-eye
- Light curtain
- Safety-edge sensing
- Open/close limit sensors

Environment

- Machine-room temperature sensor
- Humidity sensor
- Ambient-temperature measurement

Operational

- Load sensor / load cell
- Passenger-count information
- Trip count
- Start/stop count
- Travel distance

- Door-cycle count
- Running hours
- Mileage
- Drive time
- Stopping accuracy

Sensor Failure Modes

Encoder / position: drift, failure, pulse loss, signal noise, misalignment, incorrect position feedback, leveling error

Load sensor: drift, calibration error, incorrect load reading, mechanical mounting problem, wiring fault

Current sensor: calibration drift, failure, signal noise, offset error, saturation

Voltage sensor: calibration error, signal loss, noise, incorrect voltage measurement

Vibration sensor: mounting problem, noise, degradation, incorrect sampling

Temperature sensor: drift, failure, wiring fault, environmental interference

Door sensors: photo-eye contamination/misalignment, light-curtain failure, door-lock-state inconsistency, limit-sensor failure, door encoder drift

Observable Signals → Plausible Root Causes

Sensor Signal	Possible Root Cause
Floor-leveling error	Encoder / position-sensor fault
Missed floor	Encoder / position problem
Emergency stop mid-travel	Position / safety-related abnormality
Incorrect load reading	Load-sensor calibration / drift
Nuisance overload alarm	Load-sensor miscalibration
Spurious alarms	Wiring / sensor fault / noise
Inconsistent sensor readings	Sensor degradation / noise
High vibration	Bearing / traction / rope / mechanical problem
Rising temperature	Overload / friction / electrical degradation
Noisy current signal	Current-sensor / wiring / electrical issue
Position deviation	Encoder drift / failure
Unstable door-lock state	Lock sensor / contact problem
Abnormal photo-eye state	Contamination / misalignment
Unexpected door obstruction signal	Photo-eye / light-curtain / safety-edge issue

Representative chains: encoder/pulse-generator drift → floor-leveling errors/missed stops; load-sensor miscalibration → inconsistent load readings.

Putting Everything Together

The complete diagnostic model the RCA Agent can reason over spans five interconnected subsystems. They are **not isolated silos** — the research specifically requires correlation across subsystems.

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ELEVATOR
■
■■■ 1. DRIVE / MOTOR
■   ■■■ Motor / PMSM, VFD/VVVF, IGBT/inverter, DC bus, gate drivers
■   ■■■ Braking chopper/resistor, motor cables, electrical sensing
■   ■■■ Evidence: current, voltage, DC bus, temperature, speed, vibration, encoder, alarms
■   ■■■ Failures: IGBT fault, jam, winding fault, overload, overcurrent, overvoltage, overheating
■
■■■ 2. DOOR SYSTEM
■   ■■■ Door motor/operator, belt, rollers/bearings, tracks, lock/interlock
■   ■■■ Photo-eye, light curtain, safety edge, encoder, limit sensors
■   ■■■ Evidence: door current, cycle time, position, lock state, photo-eye, encoder
■   ■■■ Failures: wear, obstruction, sensor drift, lock fault, encoder fault, belt fault
■
■■■ 3. BRAKE / TRACTION
■   ■■■ Brake, brake coil, monitoring; traction sheave, ropes/belts, tension
■   ■■■ Counterweight, governor, safety gear, buffers
■   ■■■ Evidence: brake timing/torque, rope condition, tension, speed, slip, vibration, UCM
■   ■■■ Failures: brake wear, low torque, timing fault, rope degradation, slip, sheave wear
■
■■■ 4. CONTROLLER / SAFETY-CIRCUIT
■   ■■■ PLC/controller, main board, I/O, safety controller, relays, contactors
■   ■■■ Door interlocks, governor contacts, emergency stop, safety chain
■   ■■■ Evidence: event logs, fault codes, I/O states, safety-chain state, communication
■   ■■■ Failures: open safety chain, relay degradation, contact fault, I/O/comm/controller fault
■
■■■ 5. SENSORS / INSTRUMENTATION
    ■■■ Current/voltage/temperature/vibration sensors; encoders; position/leveling
    ■■■ Load sensors, door sensors, photo-eye/light curtain, environmental
    ■■■ Evidence: current, voltage, temperature, vibration, position, load, door state
    ■■■ Failures: drift, noise, calibration error, signal loss, sensor failure, misalignment
```

Cross-Subsystem Correlation Examples

These examples show why alarm correlation before RCA matters: multiple alarms can be consequences of one underlying fault.

Example 1 — Drive vs Brake

Motor Current ↑ + Brake Timing Violation + Speed Deviation

RCA considers three hypotheses:

- Drive / Motor fault
- Brake fault
- Mechanical load problem

The system must use available evidence to distinguish hypotheses — not simply conclude "overcurrent = motor fault."

Example 2 — Door + Controller

Door Closed Signal Unstable + Door Reopening + Safety Chain Open

Possible single causal chain:

Door-lock contact degradation
→ Interlock opens
→ Safety chain opens
→ Controller generates hard fault

What looks like multiple faults may actually be one causal chain.

Example 3 — Sensor → Controller → Motion

Encoder Signal Abnormal
→ Position Deviation
→ Leveling Error
→ Controller Fault / Stop Event
→ RCA traces back toward Encoder / Position Sensing

Recommended Knowledge-Base Hierarchy

For implementation, structure the knowledge base as:

Subsystem → Component → Failure Mode → Observable Signal → Evidence Pattern → Plausible Root Cause → Verification Test → Corrective Action

Example — Drive / Motor

DRIVE / MOTOR
→ IGBT Power Module
→ IGBT Short Circuit
→ Overcurrent + drive fault
→ High current + abnormal DC-bus behavior
→ IGBT failure (candidate)
→ Verify drive fault with motor-side isolation/test procedure
→ Technician repair/replacement

Example — Door System

DOOR SYSTEM
→ Roller / Bearing
→ Mechanical Wear
→ Door-cycle time ↑ + motor current ↑

→ Progressive degradation over multiple cycles
→ Worn roller/bearing (candidate)
→ Technician mechanical inspection
→ Roller/bearing replacement

Taxonomy vs Diagnosis

The "plausible root causes" in this dictionary are **candidate hypotheses**. The live RCA pipeline should only promote one to a diagnosis after evaluating the available evidence. Project documentation explicitly requires evidence-linked reasoning rather than an LLM simply guessing the cause.